CLASE 15

1. Si X e Y son variables aleatorias con densidad conjunta $f_{XY}(x;y)$ y $g:\mathbb{R}^2 \to \mathbb{R}$ es boreliana
   - $\mathbb{E}(g(X;Y)) = \iint_{\mathrm{rg}(X;Y)} g(x;y) f_{XY}(x;y)\, dxdy$
   - En particular, es válido tomar $g(x;y) = x$ y buscar usando esta integral una marginal.

2. Un accidente ocurre en un punto X uniformemente distribuido a lo largo de un camino de longitud L. Al momento del accidente, una ambulancia está situada en un punto Y también uniformemente distribuido sobre este camino. Asumiendo que X e Y son independientes, hallar la distancia esperada entre ambos y la varianza de esta distancia.

$$X;Y \overset{iid}{\sim} \mathcal{U}[0;L]$$

$d(X;Y) = |X-Y|$
$f_{XY} = f_X f_Y$ por la independencia.
$f_X(x) = \frac{1}{L}\mathbf{1}_{[0;L]}(x)$

$$\begin{aligned}
\mathbb{E}(d(X;Y)) &= \iint_{\mathrm{rg}(X;Y)} |x-y| \frac{1}{L}\mathbf{1}_{[0;L]}(x)\frac{1}{L}\mathbf{1}_{[0;L]}(y) dxdy \\
&= \frac{1}{L^2}\int_0^L \int_0^L |x-y|\, dxdy \\
&= \frac{1}{L^2}\int_0^L \int_0^L |x-y|\, dxdy
\end{aligned}$$

Vamos a definir dos regiones de integración para poder lidiar con el módulo.

$$
\begin{aligned}
\mathbb{E}\left(d\left(X;Y\right)\right) &= \frac{1}{L^2}\int_0^L\int_0^L |x-y|\,dxdy \\
&= \frac{1}{L^2}\int_0^L\left[\int_0^x (x-y)\,dy + \int_x^L -(x-y)\,dy\right]dx \\
&= \frac{1}{L^2}\int_0^L\left[\left(xy-\frac{y^2}{2}\right)\Big|_0^x + \left(xy-\frac{y^2}{2}\right)\Big|_x^L\right]dx \\
&= \frac{1}{L^2}\int_0^L\left[\left(x^2-\frac{x^2}{2}\right)+\left(xL-\frac{L^2}{2}\right)+\left(x^2-\frac{x^2}{2}\right)\right]dx \\
&= \frac{1}{L^2}\int_0^L x^2+xL-\frac{L^2}{2}dx \\
&= \frac{1}{L^2}\left(\frac{x^3}{3}+x^2\frac{L}{2}-\frac{L^2}{2}x\right)\Big|_0^L \\
&= \frac{1}{L^2}\left(\frac{L^3}{3}+\frac{L^3}{2}-\frac{L^3}{2}\right) \\
&= \frac{L}{3}
\end{aligned}
$$

$$
\mathbb{V}\left(|X-Y|^2\right) = \mathbb{E}\left(|X-Y|^2\right) - \mathbb{E}^2\left(|X-Y|\right)
$$

Necesitamos encontrar $\mathbb{E}\left(|X-Y|^2\right)$

$$
\begin{aligned}
\mathbb{E}\left(|X-Y|^2\right) &= \frac{1}{L^2}\int_0^L\int_0^L |x-y|^2\,dxdy \\
&= \frac{1}{L^2}\int_0^L\int_0^L (x-y)^2\,dxdy \\
&= \frac{1}{L^2}\int_0^L\int_0^L \left(x^2-2xy+y^2\right)dxdy \\
&= \frac{1}{L^2}\int_0^L\left(\frac{x^3}{3}-x^2y+xy^2\right)\Big|_0^L dy \\
&= \frac{1}{L^2}\int_0^L\left(\frac{L^3}{3}-L^2y+Ly^2\right)dy \\
&= \frac{1}{L^2}\left(\frac{L^3}{3}y-L^2\frac{y^2}{2}+L\frac{y^3}{3}\right)\Big|_0^L \\
&= \frac{1}{L^2}\left(\frac{L^4}{3}-\frac{L^4}{2}+\frac{L^4}{3}\right) \\
&= \frac{1}{6}L^2
\end{aligned}
$$

$$\begin{aligned}\mathbb{V}\left(|X-Y|^2\right) &= \mathbb{E}\left(|X-Y|^2\right) - \mathbb{E}^2\left(|X-Y|\right)\\ &= \frac{1}{6}L^2 - \frac{1}{9}L^2\\ &= \frac{1}{18}L^2\end{aligned}$$

3. Sean X e Y variables aleatorias continuas con función de densidad conjunta $f_{XY}(x;y) = \begin{cases} 3x & 0 \leq y \leq x \leq 1 \\ 0 & \text{toc} \end{cases}$. Hallar varianza de ambas y covarianza. Hallar el coeficiente de correlación.

$$\begin{aligned}\mathbb{E}(X) &= \iint_{\text{rg}(X;Y)} x f_{XY}(x)\, dxdy\\ &= \int_0^1 \int_0^x x \cdot 3x dydx\\ &= \int_0^1 3x^3 dx\\ &= \frac{3}{4}x^4\Big|_0^1\\ &= \frac{3}{4}\end{aligned}$$

$$\begin{aligned}\mathbb{E}\left(X^2\right) &= \iint_{\text{rg}(X;Y)} x f_{XY}(x)\, dxdy\\ &= \int_0^1 \int_0^x x^2 \cdot 3x dydx\\ &= \int_0^1 \int_0^x 3x^3 dydx\\ &= \int_0^1 3x^4 dx\\ &= \frac{3}{5}x^5\Big|_0^1\\ &= \frac{3}{5}\end{aligned}$$

$$\begin{aligned}\mathbb{V}(X) &= \mathbb{E}\left(X^2\right) - \mathbb{E}^2(X)\\ &= \frac{3}{5} - \left(\frac{3}{4}\right)^2\\ &= \frac{3}{80}\end{aligned}$$

$$\begin{aligned}
\mathbb{E}(Y) &= \iint_{\mathrm{rg}(X;Y)} y f_{XY}(y)\, dxdy \\
&= \int_0^1 \int_0^x y3xdxdy \\
&= \int_0^1 \left(\frac{y^2}{2}3x\right)\Big|_0^x dx \\
&= \int_0^1 \frac{3}{2}x^3 dx \\
&= \frac{3}{8}x^4\Big|_0^1 \\
&= \frac{3}{8}
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}(Y) &= \iint_{\mathrm{rg}(X;Y)} y f_{XY}(y)\, dxdy \\
&= \int_0^1 \int_0^x y^2 3xdxdy \\
&= \frac{1}{5}
\end{aligned}$$

$$\begin{aligned}
\mathbb{V}(Y) &= \frac{1}{5} - \left(\frac{3}{8}\right)^2 \\
&= \frac{64-45}{320} \\
&= \frac{19}{320}
\end{aligned}$$

$$\mathrm{Cov}(X;Y) = \mathbb{E}(XY) - \mathbb{E}(X)\,\mathbb{E}(Y)$$

$$\begin{aligned}
\mathbb{E}(XY) &= \iint_{\mathrm{rg}(X;Y)} xy3xdxdy \\
&= \int_0^1 \int_0^x 3x^2 ydydx \\
&= \int_0^1 \frac{3}{2}x^2y^2\Big|_0^x dx \\
&= \int_0^1 \frac{3}{2}x^4 dx \\
&= \int_0^1 \frac{3}{10}x^5 dx \\
&= \frac{3}{10}
\end{aligned}$$

$$\begin{aligned}\operatorname{Cov}(X;Y) &= \frac{3}{10} - \frac{3}{4}\cdot\frac{3}{8}\\ &= \frac{3}{160}\end{aligned}$$

Teorema 1. $\mathbb{V}\left(\sum_i X_i\right) = \sum_i \mathbb{V}(X_i) + 2\sum_{i;j} Cov(X_i;X_j)$.

3.1. Muestreo sin reemplazo.

Una población tiene elementos negros en cantidad n y verdes en cantidad v. Se extrae sin reposición una muestra de tamaño r.

Estudiemos la variable S_r : cantidad de bolitas negras extraídas cuando la extracción es de tamaño r.

La cantidad de bolitas negras extraídas tiene una distribución conocida, hipergeométrica, $S_r \sim \mathcal{H}(n+v;n;r)$.

Sea $X_k = \underset{x_i=n}{\mathbf{1}}(\boldsymbol{x})$, es decir, la variable aleatoria que indica el color de la bolita de posición i.

$$P(X_k = 1) = \frac{n}{n+v}$$

$$\begin{aligned}\mathbb{E}(X_k) &= 0\cdot P(X_k=0) + 1\cdot P(X_k=1)\\ &= P(X_k=1)\\ &= \frac{n}{n+v}\end{aligned}$$

$$\begin{aligned}\mathbb{V}(X_k) &= \mathbb{E}\left[(X-\mathbb{E}(X))^2\right]\\ &= \left(0-\frac{n}{n+v}\right)^2\cdot P(X_k=0) + \left(1-\frac{n}{n+v}\right)^2\cdot P(X_k=1)\\ &= \frac{n^2}{(n+v)^2}\frac{v}{n+v} + \left(\frac{v}{n+v}\right)^2\cdot\frac{n}{n+v}\\ &= \frac{n^2v}{(n+v)^3} + \frac{v^2n}{(n+v)^3}\\ &= \frac{nv\cancel{(n+v)}}{(n+v)^{\cancel{3}2}}\\ &= \frac{nv}{(n+v)^2}\end{aligned}$$

- Cuando $j \neq k$, $X_jX_k = 1 \Leftrightarrow x_j = x_k = n$. $P(X_jX_k=1) = \frac{n}{n+v}\frac{n-1}{n+v-1}$

$$\begin{aligned}\mathbb{E}(X_jX_k) &= 0\cdot P(X_jX_k=0) + 1\cdot P(X_jX_k=1)\\ &= \frac{n}{n+v}\frac{n-1}{n+v-1}\end{aligned}$$

$$\begin{aligned}
\operatorname{Cov}(X_j; X_k) &= \mathbb{E}(X_j X_k) - \mathbb{E}(X_j)\,\mathbb{E}(X_k) \\
&= \frac{n}{n+v}\frac{n-1}{n+v-1} - \left(\frac{n}{n+v}\right)^2 \\
&= \frac{n(n-1)(n+v)}{(n+v)^2(n+v-1)} - \frac{n^2(n+v-1)}{(n+v)^2(n+v-1)} \\
&= \frac{n^2+n^2v-n^2-nv}{(n+v)^2(n+v-1)} - \frac{n^3+n^2v-n^2}{(n+v)^2(n+v-1)} \\
&= -\frac{nv}{(n+v)^2(n+v-1)}
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}(S_r) &= \mathbb{E}\left(\sum_{i=1}^{r} X_i\right) \\
&= r \cdot \mathbb{E}(X_i) \\
&= r\frac{n}{n+v}
\end{aligned}$$

$$\begin{aligned}
\mathbb{V}(S_r) &= \sum_i \mathbb{V}(X_i) + \sum_{i;j} \operatorname{Cov}(X_i; X_j) \\
&= r\frac{nv}{(n+v)^2} - r(r-1)\frac{nv}{(n+v)^2(n+v-1)} \\
&= \frac{rnv}{(n+v)}\left\{1 - \frac{r-1}{n+v-1}\right\}
\end{aligned}$$

4. Distribución Trinomial

Definición. Se suele llamar distribución *trinomial* a la multinomial de tres categorías.

$$\boldsymbol{X} \sim \mathcal{M}(p_1; p_2; 1-p_1-p_2; n)$$

$$f_{\boldsymbol{X}}(x; y) = \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y p_3^{n-x-y}$$

- Buscar la marginal para la categoría x.

$$X \sim \mathrm{Bi}(p_1; n)$$

$$Y \sim \mathrm{Bi}(p_2; n)$$

$$\operatorname{Cov}(X; Y) = \mathbb{E}(XY) - \mathbb{E}(X)\,\mathbb{E}(Y)$$

Podemos ignorar los términos con:

- $y = 0$
- $x = 0$
- $x = n$

$$\begin{aligned} n-x-y &= n-x-y+1+1-1-1 \\ &= n-1-1-x+1-y+1 \\ &= (n-2)-(x-1)-(y-1) \end{aligned}$$

$$\begin{aligned}
\mathbb{E}(XY) &= \sum_{x=0}^{n}\sum_{y=0}^{n-x} xy \cdot \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y p_3^{n-x-y} \\
&= \sum_{x=1}^{n-1}\sum_{y=1}^{n-x} xy \cdot \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y p_3^{n-x-y} \\
&= \sum_{x-1=0}^{n-2}\sum_{y-1=0}^{n-1-(x-1)} xy \cdot \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y p_3^{n-x-y} \\
&= \sum_{x-1=0}^{n-2}\sum_{y-1=0}^{n-1-(x-1)} \cancel{x}\cancel{y} \cdot \frac{n!}{\cancel{x!}\cancel{y!}(n-x-y)!} p_1^x p_2^y p_3^{n-x-y} \\
&= p_1 p_2 n(n-1) \sum_{x-1=0}^{n-2}\sum_{y-1=0}^{n-1-(x-1)} \frac{(n-2)!}{(x-1)!(y-1)![(n-2)-(x-1)-(y-1)]!} p_1^{x-1} p_2^{y-1} p_3^{(n-2)-(x-1)-} \\
&= p_1 p_2 n(n-1) \underbrace{\sum_{x-1=0}^{n-2}\sum_{y-1=0}^{n-1-(x-1)}}_{\text{rango de la } \mathcal{M}} \underbrace{\frac{(n-2)!}{(x-1)!(y-1)![(n-2)-(x-1)-(y-1)]!} p_1^{x-1} p_2^{y-1} p_3^{(n-2)-(x-1)-}}_{\mathcal{M}(p_1;p_2;n-2)} \\
&= p_1 p_2 n(n-1) \cdot 1
\end{aligned}$$

$$\begin{aligned} \mathrm{Cov}(X;Y) &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) \\ &= p_1 p_2 n(n-1) - n p_1 n p_2 \\ &= \cancel{p_1 p_2 n^2} - p_1 p_2 n \cancel{-n^2 p_1 p_2} \\ &= -p_1 p_2 n \end{aligned}$$

5. Normal en Coordenadas Polares

$$(\rho;\theta) = g(X;Y) = \left(\sqrt{X^2+Y^2}; \mathrm{arctg}\left(\frac{Y}{X}\right)\right)$$

$$(X;Y) = g^{-1}(\rho;\theta) = (\rho\cos\theta; \rho\mathrm{sen}\theta)$$

$$F_\rho(X;Y) = \int_0^\rho \int f_{XY}(x;y) \left|J_{g^{-1}}(x;y)\right| d\theta$$

$$F_\rho(X;Y) = \int_0^\rho \int \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{\sigma^2}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{y^2}{\sigma^2}} \left|J_{g^{-1}}(x;y)\right| d\theta$$

$$J_{g^{-1}}(x;y) = \begin{vmatrix} \cos\theta & -\rho\mathrm{sen}\theta \\ \mathrm{sen}\theta & \rho\cos\theta \end{vmatrix} = \rho\cos^2\theta + \rho\mathrm{sen}^2\theta = \rho$$

$$F_\rho\left(X;Y\right)=\int_0^\rho\int_0^{2\pi}\frac{1}{2\pi\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}\rho d\theta d\rho$$

$$F_\rho\left(X;Y\right)=\int_0^\rho\int_0^{2\pi}\frac{1}{2\pi\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}\rho d\theta d\rho$$

$$F_\rho\left(X;Y\right)=\int_0^\rho\int_0^{2\pi}\frac{1}{2\pi\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}\rho d\theta d\rho$$

$$F_\rho\left(X;Y\right)=\int_0^\rho 2\pi\frac{1}{2\pi\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}\rho d\rho$$

$$F_\rho\left(X;Y\right)=\int_0^\rho \underbrace{\frac{\rho}{\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}}_{\text{densidad}}d\rho$$

- Distribución de Rayleigh.
- Si sabemos que $F_X=\int f_X dx$ podemos asegurar que la densidad de X es f_X.

$$F_\theta\left(X;Y\right)=\int_0^\theta\int_0^\infty\frac{1}{\sigma 2\pi}e^{-\frac{\rho^2}{\sigma^2}}\rho d\rho d\theta$$

$$F_\theta\left(X;Y\right)=\int_0^\theta\int_0^\infty\rho\frac{1}{2\pi\sigma}e^{-\frac{\rho^2}{\sigma^2}}d\rho d\theta$$

$z=\rho^2$
$dz=2d\rho$

$$F_\theta\left(X;Y\right)=\frac{1}{2\pi\sigma^2}\int_0^\theta\int_0^\infty e^{-\frac{z}{\sigma^2}}dzd\theta$$

$$F_\theta\left(X;Y\right)=\frac{1}{2\pi\sigma^2}\int_0^\theta -\sigma^2e^{-\frac{z}{\sigma^2}}\Big|_0^\infty d\theta$$

$$F_\theta\left(X;Y\right)=\frac{1}{2\pi\sigma^2}\int_0^\theta\left(0+\sigma^2\right)d\theta$$

$$F_\theta\left(X;Y\right)=\int_0^\theta\frac{1}{2\pi}d\theta$$

- Uniforme.

5.1. Observemos por la condición necesaria y suficiente que las variables son independientes:

$$f_R\left(\rho\right)=\frac{\rho}{\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}$$

$$f_\Theta\left(\theta\right)=\frac{1}{2\pi}$$

$$f_{R\Theta}\left(\rho;\theta\right)=f_{XY}\left(x;y\right)\left|J_{g^{-1}}\left(x;y\right)\right|=\frac{1}{2\pi\sigma^2}e^{-\frac{\rho^2}{2\sigma^2}}\rho=f_R\left(\rho\right)f_\Theta\left(\theta\right)$$